

Breath Tests in Paediatric Gastroenterology.

Three children come to clinic in the same week. A twelve-year-old girl gets bloated after drinking milk, and her lactose breath test is positive. A three-year-old boy has loose stools, and his fructose breath test is positive. A fourteen-year-old with long-standing abdominal pain had a lactulose breath test elsewhere, which was read as small intestinal bacterial overgrowth, and he has already had two courses of antibiotics. Each child has a positive breath test. The question this chapter asks is what each of those results actually proves.

The main idea is this. A breath test detects fermentation, not disease. A rise in hydrogen or methane in the breath tells you that a sugar met microbes somewhere in the gut, and roughly when. Whether that matters depends on where it happened, whether it is normal for a child of that age, and whether the child had symptoms at the time. To use breath tests well, you need to know how the gases are made, how the tests are performed, what a positive result means for each sugar, and how accurate the tests really are. We will close with the carbon thirteen urea breath test for *Helicobacter pylori*, which works on a different principle, and with what to do differently in clinic.

A short note on sources. For hydrogen and methane testing, the main document is the North American consensus published in the American Journal of Gastroenterology in 2017, written by Rezaie and colleagues. In Europe, a joint guideline from several societies, including the European Society for Paediatric Gastroenterology, Hepatology and Nutrition, called ESPGHAN, was published in 2022 with Hammer as first author, and ESPGHAN published its own position paper on breath testing in children the same year, led by Broekaert. For small intestinal bacterial overgrowth, often shortened to SIBO, we use the 2020 guideline of the American College of Gastroenterology, called ACG, written by Pimentel and colleagues. An older European document, the Rome consensus of 2009, gives the classic paediatric lactose dose. For *Helicobacter pylori* we use the 2023 joint guideline of ESPGHAN and the North American Society for Pediatric Gastroenterology, Hepatology and Nutrition, called NASPGHAN, written by Homan and colleagues.

Part one. How breath gases are made.

Human cells do not make hydrogen. Every molecule of hydrogen in the breath was made by bacteria fermenting carbohydrate. Normally almost all of that happens in the colon. When a sugar is not absorbed in the small intestine, it reaches the colon, colonic bacteria ferment it, and the hydrogen they produce diffuses into the blood and is breathed out within minutes. So a rise in breath hydrogen means that the sugar met bacteria. If the sugar is one that should have been absorbed, such as lactose, a late rise means it was not absorbed and reached the colon. If the rise comes early, it may mean that bacteria are living higher up, in the small intestine.

Methane is made by a different group of organisms called archaea, mainly one called *Methanobrevibacter smithii*. These organisms use hydrogen as fuel. They combine four molecules of hydrogen with carbon dioxide to make one molecule of methane. This has two consequences. First, a child with many methanogens may show little or no rise in hydrogen, because the hydrogen is being used up. Second, a high methane level is itself a finding. The ACG guideline now calls it intestinal methanogen overgrowth, shortened to IMO, and links it with constipation. Because archaea are not bacteria, the old term methane-positive SIBO was replaced.

Some people make no hydrogen at all on a standard test. The Rome consensus put the proportion of these hydrogen non-producers somewhere between two and forty-three percent, with lower numbers when tests run longer. A third gas, hydrogen sulphide, is made by sulphate-reducing bacteria. A three-gas device that measures it is now on the market, but no guideline mentions it and there are no accuracy data against culture.

The carbon thirteen tests are different. Here the child swallows a substance labelled with carbon thirteen, a stable, non-radioactive form of carbon. If the substance is broken down and the carbon is absorbed and burned by the body, labelled carbon dioxide appears in the breath. For *Helicobacter pylori* the substance is urea, which the bacterium splits with its urease enzyme.

The key point of this part is that the breath curve is a timeline. The same rise of twenty parts per million means malabsorption when it appears at one hundred and fifty minutes, and it is read as overgrowth when it appears at sixty minutes. Everything else in this chapter follows from that.

Part two. Doing the test properly.

The North American consensus was built with a modified Delphi process. Experts voted on twenty-eight statements, and twenty-six were accepted, which needed agreement of more than seventy percent. The level of agreement is useful to know because almost none of these statements rest on trials.

The preparation statements are these. Antibiotics should be stopped four weeks before the test, with eighty-nine percent agreement. Proton pump inhibitors should be stopped at least one week before, if the patient can manage without them, with seventy-eight percent agreement, because they speed transit and can produce false early peaks. Fermentable carbohydrates should be avoided the day before, with full agreement, and the patient should fast for eight to twelve hours, so that the baseline hydrogen is low. There should be no smoking on the day of the test and only limited exercise during it, both with full agreement. Interestingly, the experts agreed by seventy-eight percent that proton pump inhibitors do not need to be stopped for hydrogen testing. The ACG adds that people taking proton pump inhibitors who have no symptoms should not be tested for overgrowth. There was no consensus on probiotics and prebiotics. And the consensus says nothing about children. Paediatric dosing and preparation are listed as a gap in knowledge, so paediatric protocols are local.

The doses for adults are twenty-five grams of lactose, twenty-five grams of fructose, seventy-five grams of glucose, and ten grams of lactulose. For children, the Rome consensus recommends one gram of lactose per kilogram, up to twenty-five grams. There is no agreed paediatric dose of fructose, glucose or lactulose. One large paediatric series used half a gram of fructose per kilogram, up to ten grams.

The positivity criteria are simple. For lactose and fructose, a rise in hydrogen of twenty parts per million or more above baseline indicates malabsorption, and these tests should run for at least three hours. For overgrowth with glucose or lactulose, a rise of twenty parts per million or more within ninety minutes is called positive, and the test can stop at two hours. The famous double peak, an early small-bowel peak followed by a later colonic peak, is not required; the consensus authors say it has no validity. A methane level of ten parts per million or more at any time is positive. The consensus also asks that overgrowth be excluded before testing for lactose or fructose, that hydrogen, methane and

carbon dioxide be measured together, and that breath tests not be used to measure oro-caecal transit time, which is the time it takes a meal to travel from the mouth to the caecum.

Part three. Lactose: malabsorption is the global norm.

Most humans lose much of their intestinal lactase after weaning. This is called lactase non-persistence. A meta-analysis by Storhaug and colleagues, published in 2017, combined four hundred and fifty study populations, almost sixty-three thousand people from eighty-nine countries. The global prevalence of lactase non-persistence was sixty-eight percent. It ranged from about twenty-eight percent in western, southern and northern Europe to about seventy percent in the Middle East. The prevalence also depended on how it was measured: seventy-four percent by genetic testing, fifty-seven percent by hydrogen breath test, and fifty-five percent by the older lactose tolerance test. The breath test reads lower partly because of hydrogen non-producers and partly because lactase falls gradually through childhood.

How accurate is the lactose breath test? Compared with the genetic marker for non-persistence, a change in the lactase gene region called minus thirteen thousand nine hundred and ten C to T, one meta-analysis found a sensitivity of about eighty-eight percent and a specificity of about eighty-five percent. In children, an older study of three hundred and eighty-five tests found that about half were lactose malabsorbers, whether or not they produced methane. In a Spanish series of almost five hundred children with suspected functional gut symptoms, about three out of four children with the non-persistent genotype were non-absorbers, and the authors suggested genetic testing is more useful in older children.

The most important idea here is that malabsorption is not the same as intolerance. Malabsorption means the sugar was not absorbed. Intolerance means it caused symptoms. The 2022 European guideline therefore asks for a validated symptom assessment recorded at the same time as the test. Without that, a positive lactose breath test only describes physiology that the child shares with most of humanity. In most of the world, a positive result is the expected result, not a finding. The practical response is to reduce lactose rather than remove it, to re-challenge to see whether the reduction really helped, and never to restrict calcium without a plan to replace it.

Part four. Fructose: positive in most young children.

Fructose is absorbed in the small intestine by a transporter with limited capacity, and that capacity grows with age. This explains a striking finding. Jones and colleagues reviewed more than seven hundred fructose breath tests in children, using half a gram per kilogram up to ten grams. The test was positive in eighty-eight percent of infants under one year, two thirds of children aged one to five, forty percent of those aged six to ten, and twenty-seven percent of those aged ten to fifteen. The odds of a positive test fell by about eighteen percent with each year of age.

What about healthy children? Martínez-Azcona and colleagues, in 2019, tested thirty-four healthy children without symptoms and seventy-one children with chronic abdominal pain. Fructose malabsorption was found in fifty-nine percent of the healthy children and forty-one percent of those with pain. The healthy children malabsorbed more often. Some symptomatic children who tested positive did improve on a low-fructose diet, about seventy-two percent of them, but the test itself could not separate patients from healthy children.

The result also depends on the dose. In a study by Gomara and colleagues of thirty-two children with abdominal pain, none of nine children was positive after one gram of fructose, three of ten were positive after fifteen grams, and eight of thirteen were positive after forty-five grams. Nine of eleven positive children improved on a restricted diet, but there was no control group. A forty-five gram load tests the capacity of the transporter, not the presence of disease. The Rome consensus recommended fructose and sorbitol breath tests for research only, while the North American consensus accepted twenty-five grams as an adult test dose. So a positive fructose test in a young child is physiology. It cannot by itself explain abdominal pain.

Part five. Small intestinal bacterial overgrowth.

Overgrowth means that too many bacteria, usually colonic types, are living in the small intestine. It is a real problem in children with short bowel syndrome, after surgery that creates blind loops, and in severe motility disorders. The difficulty is diagnosis.

Two sugars are used. Glucose is absorbed quickly in the upper small intestine, so if hydrogen rises after glucose, bacteria must have met it before it was absorbed. The weakness is that glucose may never reach distal overgrowth. Lactulose is a sugar that humans cannot absorb at all, so it always travels to the colon. The idea is that an early rise reflects bacteria in the small intestine and a later rise reflects the colon.

How well do these tests perform against culture of fluid aspirated from the jejunum? A meta-analysis by Losurdo and colleagues in 2020, combining fourteen studies, found that the glucose test had a sensitivity of about fifty-five percent and a specificity of about eighty-three percent. The lactulose test did worse, with a sensitivity of about forty-two percent and a specificity of about seventy-one percent. An earlier systematic review by Khoshini and colleagues found very wide ranges: for glucose, sensitivity from twenty to ninety-three percent and specificity from thirty to eighty-six percent; for lactulose, sensitivity from thirty-one to sixty-eight percent and specificity from forty-four to one hundred percent. Both tests performed best after surgery that had changed the anatomy. With a sensitivity near fifty percent, a negative test does not exclude overgrowth, and with a specificity around seventy percent, a positive test often misleads.

The reason lactulose fails was shown clearly by Yu, Cheeseman and Vanner in Gut in 2011. They gave forty adults with irritable bowel syndrome a lactulose breath test together with a radioactive tracer, so that they could watch on a gamma camera when the meal reached the caecum. In eighty-eight percent of the positive tests, twenty-two of twenty-five, the tracer had already reached the caecum when hydrogen rose. The time from mouth to caecum ranged from ten to two hundred and twenty minutes between patients. The proportion of tests read as positive was thirty-five percent at ninety minutes and sixty-three percent at one hundred and eighty minutes. Normal oro-caecal transit is about eighty-three minutes, which is inside the ninety-minute window. So in many people an early lactulose peak simply tells you when the meal reached the caecum. It does not tell you that bacteria live upstream.

Glucose is not perfect either. In scintigraphy studies summarised by Lim and Rezaie in 2023, twenty-two of forty-six abnormal glucose tests turned out to be colonic fermentation, and twenty of those twenty-two patients had had upper gastrointestinal surgery that speeds delivery to the colon.

And culture, the supposed reference, has its own problems. It samples one site, it misses organisms that cannot be grown, and single-lumen aspiration catheters are contaminated in about one in five procedures. The North American experts agreed by eighty-nine percent that culture is unsatisfactory, and that if it is used the threshold should be more than one thousand colony-forming units per millilitre. In other words, accuracy figures for overgrowth tests are measured against a reference that is itself unreliable.

Part six. Breath testing in children with abdominal pain.

The paediatric data reinforce these doubts. In a randomised trial by Collins and Lin, published in 2011, seventy-five children aged eight to eighteen with chronic abdominal pain had lactulose breath tests and were randomised to rifaximin or placebo. The authors noted that earlier studies had found abnormal lactulose tests in ninety-one percent of children with chronic abdominal pain but also in thirty-five percent of healthy controls. In their trial, ninety-four percent of the rifaximin group and ninety-two percent of the placebo group had an abnormal test at baseline. Rifaximin normalised the test in only twenty percent, and symptoms did not differ between the groups. A Korean study using glucose instead found overgrowth in about twenty-one percent of sixty-eight children with functional abdominal pain disorders, a far lower figure. In paediatric functional abdominal pain, a lactulose breath test adds a label, not a treatment target.

Methane adds another layer. A fasting methane level of ten parts per million or more had a sensitivity of about eighty-six percent and a specificity of one hundred percent for methane production in data cited by the North American consensus. Methane producers are common: in one older study, sixty-two percent of sixty-five healthy volunteers produced methane. In children, rises in methane and hydrogen correlate moderately. The ACG suggests methane testing to diagnose intestinal methanogen overgrowth in constipation, as a conditional recommendation based on very low quality evidence. The practical rule is that a flat hydrogen curve without methane data cannot be interpreted, because the child may be a non-producer or a methane producer rather than truly negative.

Part seven. The carbon thirteen urea breath test.

Helicobacter pylori makes large amounts of urease. When a child swallows urea labelled with carbon thirteen, the bacterium splits it, and labelled carbon dioxide appears in the breath within about half an hour. The result is expressed as the change over baseline, and the usual cut-off is a delta over baseline above about four per mil, although some paediatric studies used two point four. The older carbon fourteen test uses a radioactive label and is not recommended in children.

A meta-analysis by Leal and colleagues, published in *Helicobacter* in 2011, combined thirty-one studies and more than four thousand children. Overall sensitivity was about ninety-six percent and specificity about ninety-six percent. In children aged six or younger, sensitivity was about ninety-five percent and specificity about ninety-three and a half percent. In children older than six, sensitivity was about ninety-seven percent and specificity about ninety-eight percent. The 2023 guideline explains that young children can have false-positive results because of their smaller distribution volume, different carbon dioxide production, and urease made by bacteria in the mouth. So the test is excellent in schoolchildren, but in a child under six a positive result deserves a second method.

The guideline is clear about how the test should be used. The first diagnosis of *Helicobacter pylori* infection in a child should be made at endoscopy with biopsy-based tests, not with a breath or stool test. The reason is that the guideline wants to know why the child has symptoms, and to culture the organism to test for antibiotic resistance, before treating. Children are not managed with the adult strategy of test and treat. The breath test, or a two-step monoclonal stool antigen test, is recommended to confirm eradication, as a strong recommendation based on high quality evidence with full agreement. It should be done six to eight weeks after the end of treatment. Proton pump inhibitors should be stopped two weeks before, and antibiotics or bismuth four weeks before. If acid suppression cannot be stopped, the child can switch to a histamine two receptor blocker and stop it two days before the test. The ESPGHAN breath testing position paper agrees: breath testing is for confirming eradication, not for diagnosing infection. In children, the breath test answers the question, is it gone? It does not answer the question, is it there?

The same position paper also recommends breath testing for overgrowth and for diagnosing and monitoring exocrine pancreatic insufficiency, using a carbon thirteen mixed triglyceride test, and it recommends against breath tests for oro-caecal transit and for coeliac disease.

Part eight. Where the guidelines agree and where they differ.

The North American and European documents agree that breath tests can be used for overgrowth, with glucose or lactulose in the North American and ACG documents and a general recommendation in the ESPGHAN paper. They agree that breath tests should not be used to measure oro-caecal transit.

They differ on what a positive carbohydrate test means. In the North American consensus a rise in hydrogen of twenty parts per million defines malabsorption. The European guideline adds that a validated symptom assessment during the test is needed before calling it intolerance. They also differ on *Helicobacter pylori*: in adults the breath test is used to test and treat, while in children endoscopy comes first and the breath test confirms cure.

The material differences are paediatric. The North American consensus does not address children and lists paediatric dosing as a gap, while the Rome consensus gives a lactose dose of one gram per kilogram up to twenty-five grams and fructose doses vary from centre to centre. Because fructose absorption depends so much on dose and age, this matters a great deal. And the North American consensus accepts twenty-five grams of fructose as a test dose, while the Rome consensus kept fructose testing for research, and healthy toddlers fail it more often than not. There are also gaps: no guideline addresses hydrogen sulphide, and only the paediatric document addresses breath tests for pancreatic insufficiency.

Part nine. What to do differently in clinic.

Here are the habits to take away. Before ordering a breath test, decide what result would change management. Check the preparation: antibiotics stopped four weeks before, laxatives and promotility drugs one week before, a low-fermentation supper and an overnight fast. Always measure methane with hydrogen. Do not order fructose breath tests in young children, because most healthy toddlers are positive, and treat a positive result as physiology. For lactose, record symptoms during the test, reduce rather than eliminate lactose, and protect calcium intake. If you must test for overgrowth, prefer glucose, save it for children with a real reason such as short bowel or previous surgery, and do not

treat functional abdominal pain on the strength of a lactulose curve. Finally, use the carbon thirteen urea breath test to confirm *Helicobacter pylori* eradication six to eight weeks after treatment, off proton pump inhibitors for two weeks, and be cautious with a positive result in a child under six.

Questions to test yourself.

Question one. What rise in breath hydrogen is considered positive, and within what time window for overgrowth? Answer: a rise of twenty parts per million or more above baseline, within ninety minutes for overgrowth.

Question two. What level of methane is considered positive? Answer: ten parts per million or more at any time during the test.

Question three. How long before a hydrogen breath test should antibiotics be stopped, and must proton pump inhibitors be stopped? Answer: four weeks for antibiotics. The North American consensus says proton pump inhibitors do not need to be stopped for hydrogen testing.

Question four. What is the global prevalence of lactase non-persistence? Answer: about sixty-eight percent.

Question five. How often was the fructose breath test positive in infants under one year in the Jones series, and in healthy children in the Martínez-Azcona study? Answer: about eighty-eight percent of infants, and about fifty-nine percent of healthy children.

Question six. In the meta-analysis against jejunal culture, what were the sensitivity and specificity of the glucose and lactulose tests? Answer: glucose about fifty-five and eighty-three percent; lactulose about forty-two and seventy-one percent.

Question seven. What did the scintigraphy study by Yu and colleagues show? Answer: in eighty-eight percent of positive lactulose tests, the meal had already reached the caecum when hydrogen rose, so the test mostly measured transit.

Question eight. When should a urea breath test be done after *Helicobacter pylori* treatment, and why is it not used for the first diagnosis in children? Answer: six to eight weeks after treatment, off proton pump inhibitors for two weeks. The first diagnosis is made at endoscopy, to find the cause of symptoms and to culture the organism for resistance.

End of chapter.